

Ex:6 CALCULATOR

AIM

To implement a simple XML-RPC-based client-server application in Python that performs basic arithmetic operations—addition, subtraction, multiplication, and division—by invoking remote procedure calls from the client to the server.

ALGORITHM

Server-Side (Server.py)

1. Import Required Module Import SimpleXMLRPCServer from the xmlrpc.server module.
2. Define Arithmetic Functions Create functions for:
 - add(x, y)
 - subtract(x, y)
 - multiply(x, y)
3. divide(x, y) (with division-by-zero check)
4. Initialize Server Create an instance of SimpleXMLRPCServer bound to localhost on port 8000.
5. Register Functions Register each arithmetic function with the server using register_function().
6. Start Server Call serve_forever() to keep the server running and listening for client requests.

Client-Side (Client.py)

1. Import Required Module Import ServerProxy from the xmlrpc.client module.
2. Connect to Server Create a proxy object pointing to the XML-RPC server at <http://localhost:8000/>.
3. Invoke Remote Methods Use the proxy to call:
 - add(5, 3)
 - subtract(10, 4)
 - multiply(6, 7)
 - divide(20, 5)
4. Display Results Print the results of each operation to the console.

OUTPUT:

```
Microsoft Windows [Version 10.0.22631.5909]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\TCS\Desktop

C:\Users\TCS\Desktop>python3 Client.py
Addition: 5 + 3 = 8
Subtraction: 10 - 4 = 6
Multiplication: 6 * 7 = 42
Division: 20 / 5 = 4.0
C:\Users\TCS\Desktop>
```

```
Microsoft Windows [Version 10.0.22631.5909]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\TCS\Desktop

C:\Users\TCS\Desktop>python3 Server.py
XML-RPC Server is running on port 8000...
127.0.0.1 - - [22/Sep/2025 16:08:35] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [22/Sep/2025 16:08:37] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [22/Sep/2025 16:08:39] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [22/Sep/2025 16:08:41] "POST / HTTP/1.1" 200 -
```

CODE:

Server.py

```
from xmlrpc.server import SimpleXMLRPCServer
```

```
def add(x, y):
    return x + y
```

```
def subtract(x, y):
    return x - y
```

```
def multiply(x, y):
    return x * y
```

```
def divide(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x / y
```

```
server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC Server is running on port 8000...")
```

```
server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")
```

```
server.serve_forever()
```

Client.py

```
import xmlrpc.client
```

```
proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")
```

```
print("Addition: 5 + 3 =", proxy.add(5, 3))
print("Subtraction: 10 - 4 =", proxy.subtract(10, 4))
print("Multiplication: 6 * 7 =", proxy.multiply(6, 7))
print("Division: 20 / 5 =", proxy.divide(20, 5))
```

RESULT:

The program has successfully completed the arithmetic operations by calling from client to the server.

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